

Ilídio Costa

Masters in Material Engineering

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<https://ilidio-costa.github.io/>

Nationality: Portuguese



EXPERIENCE

FEUP & CATEC, Porto, Portugal — *Master's Dissertation*

February 2025 - October 2025

- Led the parameter optimization for Laser Powder Bed Fusion (L-PBF) of **NiTi alloys**, successfully minimizing porosity to less than 0.1%.
- Developed and executed a Design of Experiments (DoE) utilizing Taguchi and Response Surface methodologies to reduce sample count while maximizing process efficiency.
- Conducted advanced microstructural validation using SEM, EDS, and EBSD to analyze chemical composition variations across different section thicknesses.
- Evaluated feedstock compatibility by analyzing powder feedstocks via Optical Microscopy and particle size distribution analysis

Porto Space Team, Porto, Portugal — *Advisor & Propulsion Department Leader, INVICTUS Project*

October 2024 - October 2025

- Managed the Propulsion Department budget and coordinated a multidisciplinary team to develop a hybrid propulsion motor for the INVICTUS II rocket (European Rocketry Challenge '25).
- Developed analytical models in Python to examine Hot Fire and Cold Flow data, utilizing fluid dynamics and combustion concepts to optimize engine performance.
- Established standard engineering protocols and documentation to facilitate knowledge transfer and historical context for future engineering decisions.
- Founded Project HyProSim (Hybrid Propulsion Simulator), winning "Best Poster Presentation" at the 1st Space Education Summit.
- Mentored a Bachelor's dissertation (Mechanical Engineering) on the topological optimization of a FDM avionics holding structure on nTop

Adira Addcreative, Gaia, Portugal — *Material Engineer Intern*

February 2023 - July 2023

- Executed comprehensive powder characterization for L-PBF feedstocks, analyzing flowability, morphology, and particle size distribution.
- Evaluated the degradation of metal powder over multiple print cycles and recirculations to determine material lifecycle and cost-efficiency.

INDUSTRY PROJECTS & R&D

FEUP & Palbit — *Topology Optimization for Tooling*

October 2024 - Jan 2025

- Topological optimization and Finite Element Analysis (FEA) of machining head cutter within nTop, manufactured by L-PBF.
- Poster presentation at Hi_rEV conference and publication of an article at

Professional Summary

Materials Engineer specializing in Additive Manufacturing (L-PBF & L-DED) with a strong background in process parameter optimization and powder metallurgy. Proven expertise in DfAM (nTop, Siemens NX), feedstock characterization, and post-processing heat treatments. Experienced in leading multidisciplinary engineering teams for aerospace propulsion projects.

Technical Experience

- **Additive Manufacturing:** L-PBF (Prima Sharp 150), L-DED (Process knowledge), Design for Additive Manufacturing (DfAM), Topology Optimization.

- **Materials Characterization:** SEM, EDS, EBSD, XRD, XRF, Optical Microscopy, Particle Size Distribution (ImageJ), Density (Archimedes), Mechanical Testing (Tensile, Hardness, Fatigue).

- **Software & Simulation:** nTop, Siemens NX, Materialise Magics, Abaqus, Python (Pandas, Matplotlib, pymatgen), GitHub.

- **Engineering Methodologies:** Design of Experiments (DoE), Taguchi Methods, Root Cause Analysis (Ishikawa), ISO Standards for Quality, tolerancing, and material characterization.

Languages

Portuguese - native

English - C1 CAE -Score 195/210

Metals MDPI “Topology Optimization of a Milling Cutter Head for Additive Manufacturing” (<https://doi.org/10.3390/met15070729>).

FEUP & Hypermetal — Inconel 718 L-PBF Characterization

October 2023 - Jan 2024

- Characterization from feedstock to final parts of Inconel 718 produced by L-PBF
- Published a conference article at Euro PM2025, titled “Characterization Of Inconel 718 Powder And Specimens Produced By L-PBF In Different Orientations” (<https://doi.org/10.59499/ep256768283>)

FEUP — L-DED Heat Treatment Protocol (FEUP)

October 2023

- Conducted an investigation to optimize heat treatment protocol of Inconel 718 produced by L-DED.
- Published a conference article at Euro PM2025, titled “Heat Treatments Of Inconel 718 Produced By L-DED” (<https://doi.org/10.59499/ep256767779>)

EDUCATION

FEUP, Porto, Portugal — MSc Material Engineering, Metallurgy

September 2023 - October 2025

EPMA, Lund, Sweden — Powder Metallurgy Summer School

June 2025

- Completed an intensive 5-day technical programme covering the full spectrum of Powder Metallurgy (PM) technologies, from powder production to post-sintering treatments.
- Acquired comprehensive knowledge of core and advanced manufacturing processes (Sintering, MIM, HIP, Additive Manufacturing) and analyzed material properties for steels, light metals, and functional alloys.

FEUP, Porto, Portugal — BSc Material Engineering

September 2020 - July 2023

References

Prof^a Elsa W. Sequeiros — Master’s Dissertation Supervisor;
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Msc^o Afonso Santos — Ex INVICTUS Project Lead;
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Prof^o José Costa — Bachelor’s Dissertation Supervisor;
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